Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Efficacy of Durvalumab Plus Tremelimumab in Combination With Chemotherapy Compared With Pembrolizumab in Combination With Chemotherapy in Metastatic NSCLC Patients With Non-squamous Histology Who Have Mutations and/or Co-mutations in STK11, KEAP1, or KRAS **Short Title/Acronym:** TRITON **Protocol Number:** D419ML00003 **NCT Number:** NCT06008093 **Trial Status:** Recruiting **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Durvalumab, Tremelimumab, Pembrolizumab, Pemetrexed, Carboplatin, Cisplatin (Trade Name: Platinol, ATC Code: L01XA01) **Phase of Development:** Phase III / Phase IIIb **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy of durvalumab plus tremelimumab in combination with platinum-based chemotherapy compared with pembrolizumab plus chemotherapy in terms of Overall Survival (OS) in all patients, and specifically in the subset of patients with STK11 or KEAP1 mutations/co-mutations. **Secondary Objectives:** To evaluate OS rates at 12 and 24 months, progression-free survival (PFS), objective response rate (ORR), duration of response (DoR), and overall safety and tolerability.

2.2 Background & Rationale Mutations in STK11 and KEAP1 (present in approximately 20% and 15% of patients with non-squamous metastatic NSCLC, respectively) lead to an immunosuppressive tumor microenvironment. These are often associated with inferior clinical outcomes when using standard anti-PD-(L)1 based chemo-immunotherapy, especially when co-mutated with KRAS. This study aims to determine if adding a CTLA-4 inhibitor (tremelimumab) to PD-L1 blockade (durvalumab) and chemotherapy can increase immune responses and improve overall survival for this difficult-to-treat patient population compared to the current standard of care (pembrolizumab plus chemotherapy).

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab + Chemotherapy) **Blinding:** Open-label **Randomization:** 1:1, Stratified (by mutation type [KRAS without STK11/KEAP1 capped at 33%] and tumor PD-L1 expression [$\ge 1\%$ vs $<1\%$])

3.2 Planned Enrollment & Duration Target N: 280 adult patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 04-Apr-2024 **Estimated Primary Completion:** 17-Aug-2027

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with an ECOG performance status of 0 or 1. 2. Histologically or cytologically documented Stage IV non-squamous NSCLC not amenable to curative surgery or radiation. 3. Tumors must have STK11, KEAP1, or KRAS mutations/co-mutations and lack activating EGFR mutations or ALK fusions. 4. No prior chemotherapy or systemic therapy for metastatic NSCLC (prior platinum-containing adjuvant/neoadjuvant/ definitive chemoradiation allowed if progression occurred > 6 months from end of last therapy).

4.2 Exclusion Criteria 1. Medical contraindication to platinum-based doublet chemotherapy. 2. Prior exposure to immune-mediated therapy (excluding therapeutic anti-cancer vaccines) within 6 months of randomization. 3. History of another primary malignancy (with exceptions for adequately resected non-melanoma skin cancer, curatively treated in situ disease, or malignancies curatively treated ≥ 2 years prior).

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm (T+D+CT):** Induction with Tremelimumab (75 mg) + Durvalumab (1500 mg) + Carboplatin (AUC 5/6) or Cisplatin (75 mg/m2) + Pemetrexed (500 mg/m2) every 3 weeks (Q3W) for 4 cycles. A fifth dose of Tremelimumab is given at week 16. Maintenance phase consists of Durvalumab + Pemetrexed every 4 weeks (Q4W). * **Active Comparator Arm (P+CT):** Induction with Pembrolizumab (200 mg) + Carboplatin/ Cisplatin + Pemetrexed Q3W for 4 cycles. Maintenance phase consists of Pembrolizumab + Pemetrexed Q3W.

Route of Administration: Intravenous (IV) Infusion **Duration of Treatment:** Until clinical or confirmed radiological disease progression per RECIST 1.1, unacceptable toxicity, or withdrawal (Pembrolizumab capped at 24 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care (e.g., anti-emetics for chemotherapy), hormone replacement for stable hypothyroidism, and standard diet control for existing conditions. **Prohibited:** Concurrent investigational agents, other systemic anti-cancer therapies, or systemic immunosuppressive therapies not outlined in the protocol.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Molecular Testing (STK11/KEAP1/KRAS), Baseline Imaging, Medical History
Baseline	Day 0	Randomization, First Dose Administration
Interim (Induction)	Q3W (4 Cycles)	Treatment Infusion, Safety Labs, Efficacy Assessment, AE tracking
Interim (Maintenance)	Q3W or Q4W	Maintenance Infusion, Tumor assessment per RECIST 1.1
End of Study	Follow-Up	End of treatment visit, Survival follow-up, AE resolution tracking

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Overall Survival (OS) in all participants. 2. OS in the specific subset of randomized participants with STK11 or KEAP1 mutations and/or co-mutations. **Measurement Method:** Time from randomization to death due to any cause. The measure of interest is the hazard ratio with corresponding 95% CI.

7.2 Secondary & Exploratory Endpoints * Overall survival at 12 months. * Overall survival at 24 months. * Overall survival at 12 months in the subset of randomized participants with STK11 or KEAP1 mutation and/or co-mutations. * Progression-Free Survival (PFS). * Objective Response Rate (ORR) and Duration of Response (DoR) [all investigator-assessed per RECIST v1.1]. * Safety and Tolerability (assessed via CTCAE v5.0).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous recording and grading using CTCAE v5.0. **Serious Adverse Events (SAEs):** Typical phase 3 reporting timeline (within 24 hours of site awareness). **Data Monitoring Committee (DMC):** External independent data monitoring committee to routinely review safety and efficacy signals.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy measures (OS, PFS); Safety Analysis Set for toxicities. **Sample Size Justification:** $N=280$ patients powered to evaluate OS superiority of the T+D+CT arm over the P+CT arm. **Handling of Missing Data:** Standard right-censoring mechanisms applied for survival metrics at the date of the last known contact.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring for protocol adherence, source data verification, and RECIST 1.1 tracking. **Audit Trail:** Formalized eCRF data entry, central laboratory molecular confirmation, and comprehensive safety audits.

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11. Study Identifiers & Governance

Primary ID: D419ML00003 **Registry Number:** NCT06008093 **Posting ID:** 167488459189033659 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TRITON **Official Regulatory Title:** A Phase IIIb, Randomized, Multicenter, Open-label study, to assess the Efficacy of Durvalumab Plus Tremelimumab Versus Pembrolizumab in Combination with Platinum Based Chemotherapy for First Line Treatment in Metastatic Non-Small Cell Lung Cancer Patients who have Mutations or Co mutations in STK11, KEAP1, or KRAS

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12. Clinical Framework (Metastatic NSCLC Specific)

Primary Condition: Metastatic Non-Small Cell Lung Cancer (NSCLC) - Non-squamous histology **Disease Areas:** Carcinoma, Non-Small-Cell Lung **Preferred UMLS Name:** Metastatic Non-Squamous Non-Small Cell Lung Carcinoma **Disease Classification Codes:** * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED CT IDs:** 1187425009, 1187225007, 118285006, 108369006, 1240414004 **Diagnosis Threshold:** Stage IV disease lacking EGFR/ALK alterations with confirmed STK11, KEAP1, or KRAS mutations **Medical Methodology:** Chemo-immunotherapy combination (Anti-PD-L1 + Anti-CTLA-4 + Chemotherapy) **Optimization Target:** Maximizing immune response and prolonging Overall Survival

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13. Study Procedures & Methodology

Management Pathway: First-line systemic therapy for metastatic disease **Biomarker Protocol:** Genomic screening required for STK11, KEAP1, KRAS, EGFR, ALK, and PD-L1 expression ($\ge 1\%$ vs $<1\%$) **Radiological Protocol:** Investigator-assessed tumor progression evaluations based on RECIST v1.1 criteria **Quality Audit Mechanism:** Routine sponsor oversight and safety review intervals

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14. Observation & Follow-Up Schedule

Total Duration: Up to clinical/radiological disease progression (Pembrolizumab maintenance max 24 months), followed by survival tracking **Onsite Visit Cadence:** Every 3 weeks (Q3W) during induction, shifting to Q3W or Q4W during maintenance **Imaging Frequency:** Prescribed RECIST 1.1 oncological intervals **Safety Review Interval:** Continuous monitoring per cycle and post-treatment follow-up

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]